

Radical Equations, Domains, and Decimals

Finding the domain of a radical, solving radical equations, rejecting extraneous candidates, and reporting rounded decimal answers

Directions

- Problems 1–3 warm up one skill each. From problem 4 on the skills are mixed, so read each problem and decide what it is asking for.
 - A square root is defined only where its radicand is at least 0; a radical in a *denominator* needs its radicand strictly greater than 0.
 - Squaring can create candidates the original equation never had. Test every candidate in the original equation before reporting it.
 - Where a problem asks for a decimal, round to two places and give the unit stated.
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1. Let $f(x) = \sqrt{x - 5}$.

(a) Give the domain of f as an inequality: _____

(b) Write $\sqrt{4(x - 5)}$ in the form $k\sqrt{x - 5}$: _____

2. Solve for x : $\sqrt{x - 2} = 6$

Answer: _____

3. The period of a simple pendulum, in seconds, is $T = 2\pi\sqrt{\frac{L}{9.8}}$, where L is the pendulum length in metres. Find the period of a pendulum of length $L = 2$ metres, rounded to two decimal places.

Answer: _____ s

4. Solve $\sqrt{x+2} = x$, reporting only the values that satisfy the original equation.

Answer: _____

5. Give the domain of $g(x) = \sqrt{7-2x}$ as an inequality.

Answer: _____

6. Solve for x : $\sqrt{5x-4} = \sqrt{2x+8}$

Answer: _____

7. Solve $\sqrt{2x-1} = 4.6$ for x , rounded to two decimal places.

Answer: _____

8. Solve $\sqrt{5x+6} = x$, reporting only the values that satisfy the original equation.

Answer: _____

9. Give the domain of $p(x) = \frac{1}{\sqrt{x-4}}$ as an inequality.

Answer: _____

10. Rearranging the pendulum formula gives $L = 9.8 \left(\frac{T}{2\pi} \right)^2$, with L in metres and T in seconds. Find the length of a pendulum whose period is $T = 2$ seconds, rounded to two decimal places.

Answer: _____ m

11. Solve for x : $\sqrt{x+6} = \sqrt{x} + 2$

Answer: _____

12. Synthesis. Consider the equation $\sqrt{2x+7} = x+2$.

(a) Give the domain of the radical as an inequality: _____

(b) Solve the equation, reporting only values that satisfy it: _____

(c) Squaring also produces the candidate $x = -3$. Evaluate the left side, $\sqrt{2x+7}$, at $x = -3$: _____

(d) The candidate $x = -3$ lies inside the domain you found in part (a). Explain why it is still not a solution.